

Numerical Analysis homework # 1

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I. Consider the width of the interval and the supreme of the distance of bisection method

I-a

Let the initial length of the interval $width_0$ be 2. For bisection method, the width of the interval halves each loop, that is,

$$width_{k+1} = \frac{width_k}{2}, \quad k = 0, 1, 2, \dots \quad (1)$$

Thus, by recursion, we can obtain that

$$width_n = \frac{width_0}{2^n} = 2^{-n} \times 2 = 2^{1-n} \quad n = 0, 1, 2, \dots \quad (2)$$

I-b

Let the mid point of the interval at the n th step be $m_n = \frac{a_n+b_n}{2}$. It is acknowledged that $m_0 = 2.5$. The distance of the midpoint and the root is $d = |r - m_n| = |r - \frac{a_n+b_n}{2}|$. since r is in the interval $[a_n, b_n]$, it is obvious that d will achieve its maximum value when $r = a_n$ or $r = b_n$. When $r = a_n$, we have $d = |a_n - \frac{a_n+b_n}{2}|$, while $r = b_n$, we have $d = |b_n - \frac{a_n+b_n}{2}|$. Thus, we have

$$d_{max} = |r - m_n| = |r - \frac{a_n+b_n}{2}| = \frac{b_n - a_n}{2} = \frac{width_n}{2} \quad (3)$$

When $n = 0$, d reaches its supremum, that is, $d_{max} = \frac{width_0}{2} = 1$.

II. Estimation of the relative error of bisection method

Let the accurate root be r , the estimated root at n th step be $r_n = \frac{a_n+b_n}{2}$. Then the relative error at the n th step is

$$e_n = \frac{|r - r_n|}{|r|} = \frac{|r - \frac{a_n+b_n}{2}|}{|r|} = \frac{b_n - a_n}{2r} = \frac{2^{-n}(b_0 - a_0)}{2r} = \frac{2^{-n-1}(b_0 - a_0)}{r} \quad (4)$$

Since we have $a_n \leq r \leq b_n$, we have

$$e_n = \frac{2^{-n-1}(b_0 - a_0)}{r} \leq \frac{2^{-n-1}(b_0 - a_0)}{a_0} \quad (5)$$

When n satisfies

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon - \log a_0}{\log 2} - 1 \quad (6)$$

We can obtain that

$$e_n \leq \frac{2^{-n-1}(b_0 - a_0)}{a_0} \leq \frac{2^{\frac{-\log(b_0 - a_0) + \log \epsilon + \log a_0}{\log 2} + 1 - 1} \cdot (b_0 - a_0)}{a_0} = \frac{a_0 \epsilon (b_0 - a_0)}{a_0 (b_0 - a_0)} = \epsilon \quad (7)$$

Thus, we have

$$e_n \leq \epsilon \quad (8)$$

So the relative error of bisection method is no greater than ϵ .

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III. Caculation of Newton's method

The iteration formula of Newton method is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad n \in \mathbb{N} \quad (9)$$

The derivative of $p(x)$ is $p'(x) = 12x^2 - 4x$. By using hand caculator, we have the table as follows:

x_n	$p(x_n)$	$p'(x_n)$	x_{n+1}
-1.0000000000000000	-3.0000000000000000	16.0000000000000000	(Starts iteration)
-1.0000000000000000	-3.0000000000000000	16.0000000000000000	-0.8125000000000000
-0.8125000000000000	-0.4658203125000000	11.1718750000000000	-0.7708041958041958
-0.7708041958041958	-0.0201378867202289	10.2128860824490200	-0.7688323842557603
-0.7688323842557603	-0.0000437084334135	10.1685683579878050	-0.7688280858696085

Figure 1: Newton's method iterations for $p(x) = 4x^3 - 2x^2 + 3 = 0$ starting from $x_0 = -1$.

IV. Variation of Newton's method

We have $e_{n+1} = |x_{n+1} - \alpha|$, $e_n = |x_n - \alpha|$. Through the formula of the variation of Newton's method, we can obtain that

$$e_{n+1} = |x_{n+1} - \alpha| = |x_n - \frac{f(x_n)}{f'(x_0)} - \alpha| \quad (10)$$

Since α is the true solution, that is to say, $f(\alpha) = 0$. Substitute it into equation (10) we can obtain that

$$e_{n+1} = |x_n - \frac{f(x_n) - f(\alpha)}{f'(x_0)} - \alpha| \quad (11)$$

By Mean Value Theorem, we have $f(x_n) - f(\alpha) = f'(\xi)(x_n - \alpha)$, while ξ is between x_n and α . Then the equation (11) can be

$$e_{n+1} = |x_n - \frac{f'(\xi)(x_n - \alpha)}{f'(x_0)} - \alpha| = |x_n - \alpha| |1 - \frac{f'(\xi)}{f'(x_0)}| = e_n |1 - \frac{f'(\xi)}{f'(x_0)}| \quad (12)$$

If the variation of Newton's method converges, since $\lim_{n \rightarrow \infty} x_n = \alpha$, $x_n \leq \xi \leq \alpha$ or $\alpha \leq \xi \leq x_n$, we have

$$\lim_{n \rightarrow \infty} e_{n+1} = \lim_{n \rightarrow \infty} e_n |1 - \frac{f'(\alpha)}{f'(x_0)}| \quad (13)$$

That is to say, we obtain that $\frac{e_{n+1}}{e_n} = |1 - \frac{f'(\alpha)}{f'(x_0)}|$, that is to say, $s = 1$, $C = |1 - \frac{f'(\alpha)}{f'(x_0)}|$. If we want the variation to converge, then the chosen x_0 must be close enough to α , in other words, $C < 1$.

V. Convergence of the inverse tangent function

The iteration $x_{n+1} = \tan^{-1}(x_n)$ is equivalent to $x_n = \tan(x_{n+1})$. By the property of function $y = \tan(x)$, $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, we can obtain that if $x > 0$, $\tan(x) > 0$, else if $x < 0$, $\tan(x) < 0$. That is to say, if the start point $x_0 < 0$, then $x_n < 0$ for all the $n = 0, 1, 2, \dots$, else if $x_0 > 0$, then $x_n > 0$ for all the $n = 0, 1, 2, \dots$.

And it is acknowledged that when $x \in (-\frac{\pi}{2}, 0)$, we have $\tan x < x$, when $x \in (0, \frac{\pi}{2})$, we have $\tan x > x$. That is to say, when $x_0 \in (-\frac{\pi}{2}, 0)$, $x_n = \tan x_{n+1} < x_{n+1}$, the sequence $\{x_n\}$ is monotonically increasing. By Monotonic sequence theorem, we can obtain that the iteration converge. When $x_0 \in (0, \frac{\pi}{2})$, $x_n = \tan x_{n+1} > x_{n+1}$, the sequence $\{x_n\}$ is monotonically decreasing. By Monotonic sequence theorem, we can obtain that the iteration converge.

Thus, within $(-\frac{\pi}{2}, \frac{\pi}{2})$, the iteration always converge.

VI. Convergence of the fixed-point iterations

Consider the function that $g(x) = \frac{1}{p+x}$ ($x \in [0, \frac{1}{p}]$), the continued fraction is equivalent to the fixed-point iteration $x_{n+1} = g(x_n)$. To prove the iteration converge, we can prove that the function $g(x)$ is a contractive mapping on $[0, \frac{1}{p}]$. For all $x_1, x_2 \in [0, \frac{1}{p}]$, we have

$$|g(x_1) - g(x_2)| = \left| \frac{1}{p+x_1} - \frac{1}{p+x_2} \right| = \frac{|x_1 - x_2|}{|(p+x_1)(p+x_2)|} \leq \frac{|x_1 - x_2|}{p^2} \quad (14)$$

So function $g(x)$ is a continuous contractive mapping on $[0, \frac{1}{p}]$. By the theorem Converge of contractions, the fixed-point iteration converges.

VII. Estimation of the relative error of bisection method when the initial interval includes 0

First of all, I want to answer the question "In this case, is the relative error still an appropriate measure?". The answer is no! Through the solution of II, we notice that the essential part is the estimation $|r| > a_0$, by which we limit the relative error e_n to the equation (5). However, in this case, we cannot get a lower bound for $|r|$. So the relative error may not play a role.

Instead of relative error, we choose to use absolute error $|r - \frac{a_n+b_n}{2}|$ at the nth step. By the process of bisection method, we have

$$\left| r - \frac{a_n + b_n}{2} \right| \leq \frac{|b_n - a_n|}{2} = \frac{b_0 - a_0}{2^{n+1}} \quad (15)$$

Let the absolute error $|r - \frac{a_n+b_n}{2}| \leq \epsilon$, we can transform it into $\frac{b_0-a_0}{2^{n+1}} \leq \epsilon$, that is to say,

$$n \geq \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} - 1 \quad (16)$$

VIII. Multiple zero of Newton's method

VIII-a

Let α be zero of multiplicity k of function f , then we have $f(x) = (x-\alpha)^k h(x)$, where $h(\alpha) \neq 0$ and $h(x)$ is differentiable. Let e_n be the error at nth step, then we have $e_n = x_n - \alpha$. Let's consider the relationship of errors at different steps. By Taylor theorem, we have

$$f(x_n) = f(\alpha) + f'(\alpha)(x_n - \alpha) + \frac{f''(\alpha)}{2!}(x_n - \alpha)^2 + \cdots + \frac{f^{(k)}(\alpha)}{k!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x_n - \alpha)^{k+1} \quad (17)$$

in which ξ is between x_n and α . Also, we have $f'(\alpha)$ satisfies

$$f'(x_n) = f'(\alpha) + f''(\alpha)(x_n - \alpha) + \cdots + \frac{f^{(k)}(\alpha)}{(k-1)!}(x_n - \alpha)^{k-1} + \frac{f^{(k+1)}(\zeta)}{k!}(x_n - \alpha)^k \quad (18)$$

Since $f(\alpha) = 0$ and $f^{(m)}(\alpha) = 0$ for all $m < k$, the equations (17)(18) can be written as:

$$f(x_n) = \frac{f^{(k)}(\alpha)}{k!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x_n - \alpha)^{k+1} \quad (19)$$

$$f'(x_n) = \frac{f^{(k)}(\alpha)}{(k-1)!}(x_n - \alpha)^{k-1} + \frac{f^{(k+1)}(\zeta)}{k!}(x_n - \alpha)^k \quad (20)$$

Let error error at the nth step $e_n = x_n - \alpha$, Then we can obtain that

$$\frac{f(x_n)}{f'(x_n)} = \frac{\frac{f^{(k)}(\alpha)}{k!}(x_n - \alpha)^k + \frac{f^{(k+1)}(\xi)}{(k+1)!}(x_n - \alpha)^{k+1}}{\frac{f^{(k)}(\alpha)}{(k-1)!}(x_n - \alpha)^{k-1} + \frac{f^{(k+1)}(\zeta)}{k!}(x_n - \alpha)^k} = \frac{\frac{f^{(k)}(\alpha)}{k!}e_n + \frac{f^{(k+1)}(\xi)}{(k+1)!}e_n^2}{\frac{f^{(k)}(\alpha)}{(k-1)!} + \frac{f^{(k+1)}(\zeta)}{k!}e_n} \quad (21)$$

When e_n is small enough, the equation (21) can be written as:

$$\frac{f(x_n)}{f'(x_n)} = \frac{\frac{f^{(k)}(\alpha)}{k!}e_n + O(e_n^2)}{\frac{f^{(k)}(\alpha)}{(k-1)!} + O(e_n)} = \frac{\frac{f^{(k)}(\alpha)}{k!}e_n(1 + O(e_n))}{\frac{f^{(k)}(\alpha)}{(k-1)!}(1 + O(e_n))} = \frac{1}{k}e_n(1 + O(e_n)) \quad (22)$$

Through the formula of the Newton's method, we have

$$e_{n+1} = e_n - \frac{f(x_n)}{f'(x_n)} = e_n - \frac{e_n}{k}(1 + O(e_n)) \approx \frac{k-1}{k}e_n \quad (23)$$

When e_n is small enough, through the equations (19)(20), we can obtain that

$$\frac{f(x_{n+1})}{f(x_n)} \approx \frac{\frac{f^{(k)}(\alpha)}{k!}e_{n+1}^k}{\frac{f^{(k)}(\alpha)}{k!}e_n^k} = \left(\frac{e_{n+1}}{e_n}\right)^k = \left(\frac{k-1}{k}\right)^k \quad (24)$$

Thus, when x_n is close enough to α , that is to say, e_n is small enough, we can calculate $\left|\frac{f(x_{n+1})}{f(x_n)}\right|$, which will stabilize to a value, let it be L then we can solve the equation $\left(\frac{k-1}{k}\right)^k = L$, finally, we get k , which will help us to detect the multiple zero.

VIII-b

Since r is a zero of multiplicity k of the function f , we can obtain that:

$$f(x) = (x - r)^k h(x) \quad (25)$$

where $h(\alpha) \neq 0$ and $h(x)$ is differentiable. While $f'(x) = k(x - r)^{k-1}h(x) + (x - r)^k h'(x)$, we can obtain that

$$\frac{f(x)}{f'(x)} = \frac{(x - r)h(x)}{kh(x) + (x - r)h'(x)} \quad (26)$$

Through the modification of Newton's method, we have:

$$e_{n+1} = x_{n+1} - r = x_n - r - k \frac{(x_n - r)h(x_n)}{kh(x_n) + (x_n - r)h'(x_n)} = e_n - k \frac{e_n h(x_n)}{kh(x_n) + e_n h'(x_n)} \quad (27)$$

that is to say:

$$e_{n+1} = e_n \left(1 - \frac{1}{1 + \frac{e_n}{k} \frac{h'(x_n)}{h(x_n)}}\right) = \frac{\frac{e_n^2}{k} \frac{h'(x_n)}{h(x_n)}}{1 + \frac{e_n}{k} \frac{h'(x_n)}{h(x_n)}} \quad (28)$$

When $n \rightarrow \infty$, we have $e_n \rightarrow 0$, $\frac{h'(x_n)}{h(x_n)} \rightarrow \frac{h'(r)}{h(r)}$, then we can obtain that:

$$\lim_{n \rightarrow \infty} \frac{|e_{n+1}|}{|e_n|^2} = \frac{h'(r)}{kh(r)} \quad (29)$$

Which implies that quadratic convergence in Newton's iteration is restored.